MIL-R-10265D
17 January 1974
SUPERSEDING
MIL-R-10265C
15 April 1965

MILITARY SPECIFICATION

REGULATOR, ENGINE GENERATOR; 28-VOLT DIRECT CURRENT, VIBRATING TYPE

THIS SPECIFICATION IS APPROVED FOR THE USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

1. SCOPE

1.1 Scope. This specification covers vibrating type generator regulators incorporating devices to control reverse current and to regulate voltage and current. The regulator is inclosed in a single housing for use in regulating 28-volt direct current generators (see 6.1).

1.2 Classification. Regulators shall be of the following grades as specified (see 6.2 and 6.3):

Grade A	-	Waterproof.
Grade B	-	Nonwaterproof.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date in invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-E-489	-	Enamel; Alkyd Gloss (for Exterior and Interior Surfaces).

Military

MIL-P-514	-	Plates, Identification, Transportation - Data and Blank.
MIL-B-11188	-	Batteries, Storage: Lead-Acid.
MIL-G-12604	-	Generator, Engine Accessory: Direct-Current, 28 Volt Nominal.
MIL-E-13856	-	Electrical Components for Automotive Vehicles; Waterproofness and Resistance to Temperature Cycling Tests, General Requirements for.

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STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MS-13803	-	Regulator, Voltage and Current (Generator, Engine Accessory, 24-volt, 40-Amp).
MS-13805	-	Regulator, Voltage and Current (Generator, Engine Accessory, 24-Volt, 18-Amp).

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Service.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 First Article. When specified (see 4.2 and 6.2), three samples of the regulators to be furnished under this specification shall be produced prior to the manufacture of these items in production quantity. Under the surveillance of Government representatives, designated by the contracting officer, the manufacturer shall subject these regulators to the first article testing specified herein to determine conformance to the quality assurance provisions of this specification. The government representatives shall determine which of the samples shall undergo which sequence of first article testing (see 4.2). The samples shall be fully representative of the regulators to be supplied from production facilities and tooling.

3.2 Materials. Materials shall be as specified herein and on applicable drawings, standards and specifications.. When the material used is not specifically covered in any of the referenced documents, it shall withstand the temperature range encountered within the component for which it is intended for use without any change in physical or chemical properties resulting in calibration or operation to fall outside limits specified herein.

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3.3 Design and construction. In order to assure mechanical and electrical interchangeability, regulators shall be fabricated in conformance with applicable MS standards or drawings (see 6.2).

3.3.1 Rating. The amperage rating shall be as specified on the applicable drawing or standard.

3.3.2 Standard parts. Military standard parts shall be used whenever possible.

3.3.3 Threaded parts. All screw threads shall conform to Handbook H28. The form, number per inch and class of threads shall be as specified on applicable standards or drawings.

3.3.4 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. Where such contact is not necessary to complete an electrical circuit, but is otherwise unavoidable, surfaces shall be insulated.

3.3.5 Fungus resistance. Materials used shall conform to the fungus resistance requirements of requirement 4 of MIL-STD-454.

3.3.6 Polarity. Unless otherwise specified on the applicable drawing or standard, the regulators shall be designed to operate with a negatively-grounded electrical system.

3.3.7 Provisions for servicing. Servicing of regulators, including replacement of major assemblies, shall be possible without the use of special tools.

3.4 Performance.

3.4.1 Reverse current control.

3.4.1.1 Closing voltage. The cut out relay of the regulator shall close the circuit between the battery and the generator. The closing voltage shall be 26 $\pm$ 1 volts.

3.4.1.2 Opening current. The cut out relay of the regulators shall open the circuit between the battery and the generator when the reverse current reaches not less than 15 percent nor more than 30 percent of rated current.

3.4.2 Voltage regulation. The generator regulator shall establish and maintain the output voltage at 28.5 $\pm$ 1.5 volts. In addition, the voltage regulator shall reestablish and maintain the output voltage at 28.5 $\pm$ 1.5 volts within 0.1 seconds of any load change and shall not develop sustained oscillation. The peak of ripple voltage to average direct current (DC) voltage shall not exceed two volts.

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3.4.3 Current regulation. The regulator shall regulate the current output of the generator within the limits 100 ± 10 percent of the generator rated output current.

3.4.4 Endurance. The generator regulator shall withstand 700 hours of operation under the load, speed and temperature conditions specified herein without servicing, adjustment or replacement of parts. The regulator shall subsequently meet the requirements of 3.4.1, 3.4.2 and 3.4.3.

3.4.5 High voltage. When subjected to a voltage of 500 volts root mean square at a frequency of 60 hertz for the length of time specified herein, the regulator shall evidence no smoking, sparking, arcing, flashover, burning or charring of the insulation, conducting cables, connectors, insulated terminals, windings and other conducting parts. The regulator shall subsequently meet the requirements of 3.4.1, 3.4.2 and 3.4.3.

3.4.6 Vibration. When subjected to vibration of the frequences and amplitudes and for the length of time specified herein, the regulator shall evidence no chattering of the cutout relay contact points, loosening of parts, abrasion, breakage or other physical damage.

3.4.7 Shock resistance. The generator regulator shall withstand the sawtooth shock pulses specified herein with no evidence of cutout relay contact points closing; no breakage, loosened or distorted parts or other physical damage. The regulator shall subsequently meet the requirements of 3.4.1, 3.4.2 and 3.4.3.

3.4.8 Extreme temperature resistance. Following stabilization at the temperatures specified herein, the regulator shall meet the requirements of 3.4.1, 3.4.2 and 3.4.3, except that the output voltage (see 3.4.2) shall be 27.5 ± 1.5 volts at plus 160°F.

3.4.9 Corrosion resistance. The regulator shall withstand salt spray corrosion for the length of time herein specified and shall evidence no corrosion that would adversely affect performance. The regulator shall subsequently meet the requirements of 3.4.1, 3.4.2 and 3.4.3.

3.4.10 Waterproofness. Grade A regulators shall conform to the waterproofness requirements specified for type II, class 1, components of MIL-E-13856.

3.5 Finish. Every exterior metal surface, other than the electrical connectors, shall be painted. All painted surfaces shall be primed and subsequently finish painted with gloss enamel.

3.5.1 Film thickness. Prime and finish coats shall have a combined minimum thickness of one mil.

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3.5.2 Enamel. Finish coat enamel shall be black-gloss enamel conforming to TT-E-489.

3.6 Identification of product.

3.6.1 Marking. The regulators shall be marked in accordance with MIL-STD-130.

3.6.2 Nameplate. A nameplate conforming to MIL-P-514 shall be attached to the regulator housing in the manner specified on the applicable standard or drawing. The legend contained on the plate shall include the following as a minimum:

Regulator, Generator
AMPS, 28 Volts DC
Specification - MIL-R-10265
Manufacturer's Identification
Manufacturer's Part Number
Serial Number
Military Part Number
Federal Stock Number
US

3.7 Workmanship. Workmanship shall conform to the requirements of requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First Article inspection. When required (see 3.1 and 6.2), samples shall consist of three regulators. First article inspection and testing shall be performed by the contractor under Government surveillance and shall consist of examination for defects specified in 4.3.2.2 and performance of tests specified in table I, in the order listed.

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Table I. Order of First Article Testing

Test	Requirement	Paragraph No.	Specimen 1	2	3
Reverse current control	3.4.1	4.5.1	X	X	X
Voltage regulation	3.4.2	4.5.2	X	X	X
Current regulation	3.4.3	4.5.3	X	X	X
Extreme temperature	3.4.8	4.5.8		X	X
Endurance	3.4.4	4.5.4	X		
Vibration	3.4.6	4.5.6		X	
Corrosion	3.4.9	4.5.9		X	
Shock	3.4.7	4.5.7			X
Waterproofness (Grade A only)	3.4.10	4.5.10		X	X
High voltage	3.4.5	4.5.5		X	X

4.2.1 Failure. Failure of a first article sample to pass any of the examinations or tests specified herein, may be cause for the Government's refusal to witness additional examinations of tests until the faults revealed have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. A lot shall consist of all regulators of one grade or rating, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for testing. A representative sample shall be selected in accordance with inspection level S-3 of MIL-STD-105 from each lot that has passed the examination specified in 4.3.2.2.

4.3.2 Examination.

4.3.2.1 Acceptable quality level. Each regulator selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

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4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as follows:

Major defects (1.0 AQL)	Examination Method
Incorrect dimensions affecting interchangeability (see 3.3)	SIE*

Minor defects (2.5 AQL)	Examination Method
Incorrect dimensions not affecting interchangeability (see 3.3)	SIE*
Improper finish (see 3.5)	Visual
Improper identification of product (see 3.6)	Visual
Poor workmanship (see 3.7)	Visual

*SIE - Standard inspection equipment

4.3.3 Quality conformance tests. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in 4.5.1, 4.5.2, 4.5.3 and 4.5.10.2 using an AQL of 6.5 percent defective.

4.3.4 Rejection and retest. Regulators that have been rejected may be reworked or have parts replaced to correct defects and be resubmitted for tests. Before resubmitting, detailed information or data concerning the rejection and the action taken to correct the defects shall be furnished to the Government inspector.

4.4 Control tests.

4.4.1 Sampling. Control test samples shall be selected at the rate of one of each 500 units produced, except that not less than one nor more than three units shall be selected in any 30-day period. Regulators selected shall be examined for the defects specified in 4.3.2.2, and subsequently tested as specified in 4.5.1, 4.5.3, 4.5.6, 4.5.7 and 4.5.10.1 in the order listed.

4.4.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the examination or test have been corrected.

4.5 Test procedure. Unless otherwise specified herein, tests shall be conducted under the following conditions:

(a) Ambient air temperature shall be $77^{\circ} \pm 15^{\circ}F$.
(b) Regulators shall be stabilized and adjusted to set voltage before being subjected to tests (see 6.4).

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(c) The generator used in the test circuit shall have the same rating as the regulator being tested and shall conform to MIL-G-12604.

(d) The battery shall consist of two 12-volt batteries conforming to type 2HN of MIL-B-11188. The two batteries shall be connected in series. Batteries shall be fully charged and shall be kept at room temperature (see 6.4).

(e) Test equipment accuracy shall be such as to permit measurement within 10 percent of product or test specification tolerance. Calibration of test equipment shall be conducted at intervals sufficient to establish required accuracy. Records of calibration shall be made available to the Government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the Government. Instruments such as voltmeters, ammeters and thermometers shall be accurate to within 0.25 percent of full scale reading for preproduction testing and shall be accurate to within 1.0 percent of full scale reading for all other testing.

4.5.1 Reverse current control.

4.5.1.1 Closing voltage. To determine conformance of regulators to 3.4.1.1, the generator shall be driven with the speed gradually being increased until the cutout relay closes the circuit. The process shall be repeated five times with the closing voltage recorded each time.

4.5.1.2 Opening current. To determine conformance of regulators to 3.4.1.2, the generator shall be operated at a speed sufficient to deliver charging current to the battery. The cutout relay shall have 25 $\pm$ 1 volts impressed on its voltage winding. The generator speed shall then be decreased gradually until the cutout relay opens the circuit. The process shall be repeated five time with the opening current recorded each time.

4.5.2 Voltage regulation. To determine conformance to 3.4.2, the test specimen shall be tested as follows: The battery is to remain connected with the generator and regulator throughout the test. Additional load by resistance is to be adjusted so that the total load of battery and resistance will establish the current at 80 percent rated load current. During the test, the resistance load shall be applied and removed suddenly.

(a) With generator operating at minimum speed for regulation (see 6.4), with no additional resistance load, record the voltage.

(b) Apply the resistance load required to establish 80 percent rated load current and record the voltage.

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(c) Remove the resistance load, record the voltage, increase the generator speed to middle speed (approximately midway between minimum speed for regulation and maximum rated speed), and record the voltage.
(d) Apply the resistance load required to establish 80 percent rated load current and record the voltage.
(e) Remove the resistance load, record the voltage, increase the generator speed to maximum rated speed, and record the voltage.
(f) Apply the resistance load required to establish 80 percent rated load current and record the voltage.
(g) Without changing the resistance load, reduce the generator speed to minimum speed for regulation and record the voltage.
(h) Remove the resistance load and record the voltage.
(i) With no resistance load, increase the generator speed to maximum rated speed and record the voltage.

4.5.3 Current regulation. With the generator producing maximum rated voltage at middle speed, with no battery connected, the resistance load shall be adjusted to establish 80 percent of rated load current. The load current shall be increased gradually by adjusting the resistance load until the voltage at the battery terminal of the test sample has dropped to 25 $\pm$ 0.5 volts. The load current at 25 $\pm$ 0.5 volts shall be recorded to determine conformance to 3.4.3.

4.5.4 Endurance. The regulator shall be electrically connected with a generator and a battery and cycled (see 4.5.4.1) for 700 hours as follows:

(a) 3500 cycles at minus 65° $\pm$ 5°F. (Prior to cycling, the regulator shall be maintained at minus 65°F $\pm$ 5°F for two hours.)
(b) 10,500 cycles at plus 77° $\pm$ 15°F. (Prior to cycling, the regulator shall be maintained at plus 77° $\pm$ 15°F for two hours.)
(c) 3500 cycles at plus 160° $\pm$ 5°F. (Prior to cycling the regulator shall be maintained at plus 160° $\pm$ 5°F for two hours.)

During one-half of the test time at each temperature, the regulator shall operate with a load 20 $\pm$ 5 percent of rated output, while during the other half it shall operate with a load adjusted to produce a voltage of 25 $\pm$ 1 volts. The specimen shall subsequently be tested as specified in 4.5.1, 4.5.2 and 4.5.3 to determine conformance to 3.4.4, (see 6.4).

4.5.5 High voltage. Conducting cables, connectors, insulated terminals, windings, and other conducting parts of the regulator (except capacitors, parts which include capacitors within sealed case, or rectifiers) shall be subjected to conductor-to-ground and conductor-to-conductor tests. The voltage shall be 500 volts root mean square at a frequency of 60 hertz and shall be applied after appropriate grounds are disconnected and after preliminary low-voltage continuity and ground tests. The voltage shall be applied for one minute for preproduction and control testing, and one second for acceptance testing. Close observation shall be maintained during test to determine conformance to 3.4.5. The regulator shall subsequently be tested in accordance with 4.5.1, 4.5.2 and 4.5.3 as applicable.

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4.5.6 Vibration test. To determine conformance to 3.4.6, the regulator, including all mounting shown on the applicable MS standard, shall be mounted on a vibrating machine to simulate actual mounting in use. The regulators shall then be subjected to simple harmonic vibration in the direction of each of the three major axes through frequency ranges and travels and time intervals as follows:

Frequency range (Cycles per second)	Total travel (Inches, maximum)		Time (Hours)
	Non-operating	Operating	
10-20	0.100	.025	2
20-30	0.050	.015	2
30-50	0.015	.003	2

The vibrating machine shall be adjusted to yield the maximum travel at some point within the frequency range. The entire frequency range shall be accomplished each minute for each specified travel. During the first hour of each 2-hour period, the generator regulator shall be electrically inoperative and during the last hour it shall be connected and operated in accordance with 4.5.1, 4.5.2 and 4.5.3. After the non-operating portion and during the operating portion of each test, careful inspection shall be maintained to determine conformance to 3.4.6.

4.5.7 Shock test. To determine conformance to 3.4.7, the regulator shall be tested in accordance with method 213 of MIL-STD-202 using test condition I. The regulator, including all mounting shown on the applicable MS standard, shall be mounted to simulate actual mounting in use. The regulator shall subsequently be subjected to the tests specified in 4.5.1, 4.5.2 and 4.5.3.

4.5.8 Extreme temperature resistance. To determine conformance to 3.4.8, the regulator shall be subjected to the ambient air temperature for the specified stabilization period and subsequently subjected to tests as specified in table II.

Table II. Extreme Temperature Calibration Tests

Temperature	Temperature Stabilization Period	Tests after Stabilization
80° ± 5°F.	24 hours	None
65° ± 5°F.	24 hours	4.5.1, 4.5.2, 4.5.3
160° ± 5°F.	24 hours	4.5.1, 4.5.2, 4.5.3
225° ± 5°F.	24 hours	4.5.1, 4.5.2, 4.5.3

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4.5.9 Corrosion-resistance. The regulator shall be subjected to the salt spray corrosion-resistance test specified in method 101 of MIL-STD-202. Grade A regulators (see 1.2) shall be tested for 300 hours sealed. Grade B regulators (see 2.1) shall be tested for 50 hours open. Hermetically-sealed component parts shall remain sealed during the test. When applicable, the regulator contacts shall be cleaned and the regulator recalibrated. A 20 percent salt-solution concentration shall be used. The specimen shall subsequently be tested in accordance with 4.5.1, 4.5.2 and 4.5.3 to determine conformance to 3.4.9.

4.5.10 Waterproofness. Grade A regulators shall be subjected to applicable waterproofness tests for components classified in MIL-E-13856 as type II, class 1.

4.5.10.1 Preproduction and control. To determine conformance to 3.4.10, grade A regulators shall be subjected to the preproduction and control waterproofness test specified in MIL-E-13856.

4.5.10.2 Quality conformance. To determine conformance to 3.4.10, grade A regulators shall be subjected to the quality conformance test specified in MIL-E-13856.

4.6 Inspection of preparation for delivery.

4.6.1 Materials and processes. The Government inspector shall at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to requirements of section 5. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The regulators covered by this specification are intended for use in 28-volt DC nominal electrical systems equipped with charging generators. The regulators are intended for use on off-road engineering vehicles and inductrical applications.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Grade of regulator required (see 1.2).
(c) First article testing, if required (see 3.1 and 4.2).
(d) Applicable drawing or Military standard (see 3.3).
(e) Selection of applicable levels of preservation and packaging (see 5.1).

6.3 Change of classification. This revision deletes reference to types I and II in the classification of regulators. Solid state regulators, which were type I, are now covered by other military specifications.

6.4 Definitions.

6.4.1 Minimum rated speed. The term as used in this specification applies to "generator minimum rated speed". Generator minimum rated speed is the lowest speed at which generator will deliver rated current at 28.5 ± 0.1 volts. Rated current for the generator is the current indicated on the applicable MS standard or drawing.

6.4.2 Minimum speed for regulation. The minimum speed for regulation is a generator speed 500 RPM greater than the minimum rated speed.

6.4.3 Cycle. A cycle shall consist of varying the generator speed from maximum rated speed, down to low speed (see 4.5.4.2) and then returning to maximum rated speed. The generator speed shall be varied in such a manner that the cycling time above (or below) middle rated speed will be 50 ± 10 percent fof the total cycling time.

6.4.4 Low speed. The speed low enough to cause the cutout relay to open.

6.4.5 Set voltage. Set voltage is the voltage to which the regulator output is adjusted with the generator operating at average speed and with no load except the battery. Setting is accomplished after the regulator has been stabilized.

6.4.6 Regulator, stabilized. A stabilized regulator is defined as a regulator which is being operated at minimum speed for regulation with resistance and battery adjusted so as to extablish 80 percent of rated load current at 28 volts; and which when so operated experiences a temperature change of less than 2°F over a 15 minute period.

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6.4.7 Fully charged battery. For the purposes of the test procedures in this specification, a fully charged battery is defined as a battery having a current consumption of not more than four amperes at 28 volts at room temperature (77° ± 15°F).

Custodians:
Army - AT
Navy - YD

Preparing activity:
Army - AT

Review activities:
Army - AT, ME
Defense Supply Agency - CS

Project No. 2920-0287

User activities:
Army - EL
Navy - YD, MC

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